FACILITATION GUIDE

LESSON TITLE: Operating Systems Performance Examination

LESSON DESIGNATOR: DS.24.03

TOTAL LESSON TIME: 5 Hours

LEARNING OBJECTIVES:

ELOs:

1. With the aid of reference and given the proper equipment and software, configure group policy objects, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002aa).
2. With the aid of reference and given the proper equipment and software, maintain organizational units, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ac).
3. With the aid of reference and given the proper equipment and software, navigate a client operating server, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ad).
4. With the aid of reference and given the proper equipment and software, configure basic server configurations, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ae).
5. With the aid of reference and given the proper equipment and software, maintain active directory groups, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002af).
6. With the aid of reference and given the proper equipment and software, connect a server to a network, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ag).
7. With the aid of reference and given the proper equipment and software, link group policy objects to active directory, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ah).
8. With the aid of reference and given the proper equipment and software, join a domain, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ai).
9. With the aid of reference and given the proper equipment and software, monitor active directory replication, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ak).
10. With the aid of reference and given the proper equipment and software, backup active directory data, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002al).
11. With the aid of reference and given the proper equipment and software, add a client to an active directory domain, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002am).
12. With the aid of reference and given the proper equipment and software, restore active directory data backups, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002an).
13. With the aid of reference and given the proper equipment and software, test server connectivity, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ao).
14. With the aid of reference and given the proper equipment and software, troubleshoot server connectivity, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ap).

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15. With the aid of reference and given the proper equipment and software, update a server operating system, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002aq).
16. With the aid of reference and given the proper equipment and software, verify client connectivity to the network, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002at).
17. With the aid of reference and given the proper equipment and software, test client connectivity to the network, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002au).
18. With the aid of reference and given the proper equipment and software, use active directory administrative tools, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002aw).
19. With the aid of reference and given the proper equipment and software, navigate an operating system, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002az).
20. With the aid of reference and given the proper equipment and software, deploy operating systems, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002b).
21. With the aid of reference and given the proper equipment and software, install an active directory domain server, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bb).
22. With the aid of reference and given the proper equipment and software, create a domain, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002be).
23. With the aid of reference and given the proper equipment and software, install a domain name server, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bf).
24. With the aid of reference and given the proper equipment and software, maintain active directory objects, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bg).
25. With the aid of reference and given the proper equipment and software, force implement group policy objects, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bh).
26. With the aid of reference and given the proper equipment and software, change a servers computer name, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bi).
27. With the aid of reference and given the proper equipment and software, enforce active directory domain policies on a client, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bj).
28. With the aid of reference and given the proper equipment and software, utilize a naming standard, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bk).
29. With the aid of reference and given the proper equipment and software, connect a workstation client to a network, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bl).
30. With the aid of reference and given the proper equipment and software, change a workstation clients computer name, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bm).
31. With the aid of reference and given the proper equipment and software, use server monitoring tools, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bo).

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32. With the aid of reference and given the proper equipment and software, restore a server operating system, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bs).
33. With the aid of reference and given the proper equipment and software, monitor a domain name server, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bt).
34. With the aid of reference and given the proper equipment and software, remove client from the domain, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bw).
35. With the aid of reference and given the proper equipment and software, create domain name server records, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002by).
36. With the aid of reference and given the proper equipment and software, create organizational units, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bz).
37. With the aid of reference and given the proper equipment and software, configure operating systems, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002c).
38. With the aid of reference and given the proper equipment and software, create user objects, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ca).
39. With the aid of reference and given the proper equipment and software, create security groups, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002cb).
40. With the aid of reference and given the proper equipment and software, promote a domain controller, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002cc).
41. With the aid of reference and given the proper equipment and software, assign object permissions, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002cd).
42. With the aid of reference and given the proper equipment and software, backup active directory data, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002ci).
43. With the aid of reference and given the proper equipment and software, troubleshoot operating systems, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002d).
44. With the aid of reference and given the proper equipment and software, configure domain name services conditional forwarder records, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002dy).
45. With the aid of reference and given the proper equipment and software, deploy a messaging server, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002el).
46. With the aid of reference and given the proper equipment and software, configure a messaging server, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002em).
47. With the aid of reference and given the proper equipment and software, configure a messaging client, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002en).
48. With the aid of reference and given the proper equipment and software, deploy domain services, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002f).
49. With the aid of reference and given the proper equipment and software, install messaging server prerequisite software, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fe).

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50. With the aid of reference and given the proper equipment and software, configure messaging server administrator accounts, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002ff).
51. With the aid of reference and given the proper equipment and software, connect a messaging server to the network, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fh).
52. With the aid of reference and given the proper equipment and software, prepare messaging server organizations, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fi).
53. With the aid of reference and given the proper equipment and software, prepare domain schemas for messaging servers, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fj).
54. With the aid of reference and given the proper equipment and software, prepare an active directory forest for messaging services, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fk).
55. With the aid of reference and given the proper equipment and software, verify messaging server installation, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fl).
56. With the aid of reference and given the proper equipment and software, perform a gui-based messaging server installation, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fm).
57. With the aid of reference and given the proper equipment and software, install messaging server roles, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fn).
58. With the aid of reference and given the proper equipment and software, access messaging administrative tools, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fo).
59. With the aid of reference and given the proper equipment and software, configure messaging administrative tools, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fp).
60. With the aid of reference and given the proper equipment and software, create messaging users, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fq).
61. With the aid of reference and given the proper equipment and software, assign domain user mailboxes, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fr).
62. With the aid of reference and given the proper equipment and software, configure mail-enabled groups, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fs).
63. With the aid of reference and given the proper equipment and software, enable external mail flow, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002ft).
64. With the aid of reference and given the proper equipment and software, configure messaging recipients, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fu).
65. With the aid of reference and given the proper equipment and software, configure send connectors, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fv).
66. With the aid of reference and given the proper equipment and software, configure receive connectors, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fw).

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67. With the aid of reference and given the proper equipment and software, configure mailbox databases, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fx).
68. With the aid of reference and given the proper equipment and software, migrate mailboxes, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fy).
69. With the aid of reference and given the proper equipment and software, configure domain name services mail exchanger records, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fz).
70. With the aid of reference and given the proper equipment and software, configure domain services, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002g).
71. With the aid of reference and given the proper equipment and software, access messaging applications, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002ga).
72. With the aid of reference and given the proper equipment and software, navigate messaging applications, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002gb).
73. With the aid of reference and given the proper equipment and software, configure messaging clients, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002gc).
74. With the aid of reference and given the proper equipment and software, send emails, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002gd).
75. With the aid of reference and given the proper equipment and software, import data from personal folders to messaging clients, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002ge).
76. With the aid of reference and given the proper equipment and software, navigate OS registry, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002gf).
77. With the aid of reference and given the proper equipment and software, restore domain services, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002h).
78. With the aid of reference and given the proper equipment and software, mount server installation media, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002t).
79. With the aid of reference and given the proper equipment and software, set up server operating system installation, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002u).
80. With the aid of reference and given the proper equipment and software, select server operating system editions, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002v).
81. With the aid of reference and given the proper equipment and software, select the server operating system installation option, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002w).
82. With the aid of reference and given the proper equipment and software, install a workstation client, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002x).
83. With the aid of reference and given the proper equipment and software, install server roles, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002y).
84. With the aid of reference and given the proper equipment and software, install server features, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002z).

85. Without the aid of reference, identify in writing the high-availability solutions, in accordance with ISBN: 978-1-119-23205-6. (0671-OPER-1002a).
86. Without the aid of reference, migrate active database, in accordance with ISBN: 978-1-119-23205-6. (0671-OPER-1002b).
87. Without the aid of reference, unmount active database, in accordance with ISBN: 978-1-119-23205-6. (0671-OPER-1002c).
88. Without the aid of reference, backup active database, in accordance with ISBN: 978-1-119-23205-6. (0671-OPER-1002d).
89. Without the aid of reference, gracefully shut down messaging server, in accordance with ISBN: 978-1-119-23205-6. (0671-OPER-1002e).
90. Without the aid of reference, update messaging server, in accordance with ISBN: 978-1-119-23205-6. (0671-OPER-1002f).
91. Without the aid of reference, use messaging tools to track email flow, in accordance with ISBN: 978-1-119-23205-6. (0671-OPER-1002g).
92. Without the aid of reference, update messaging service's prerequisite software, in accordance with ISBN: 978-1-119-23205-6. (0671-OPER-1002h).
93. Without the aid of reference, run messaging server health-check commands, in accordance with ISBN: 978-1-119-23205-6. (0671-OPER-1002i).
94. Without the aid of reference, update server operating system, in accordance with vendor's best practices. (0671-OPER-1002j).
95. Without the aid of reference, backup server operating system, in accordance with vendor's best practices. (0671-OPER-1002k).
96. Without the aid of reference, update domain service, in accordance with vendor's best practices. (0671-OPER-1002l).
97. Without the aid of reference, restart domain service, in accordance with vendor's best practices. (0671-OPER-1002m).
98. Without the aid of reference, validate availability of domain service, in accordance with vendor's best practices. (0671-OPER-1002n).
99. Without the aid of reference, download the current Security Technical Implementation Guides (STIGs), in accordance with USMC ECSM Library. (0671-OPER-1002o).
100. Without the aid of reference, perform a Security Technical Implementation Guide scan validation, in accordance Security Center Administrators Guide. (0671-OPER-1002p).

REFERENCES:

Reference Title	Publication ID
Defense Information Systems Agency (DISA) Security Technical Implementation Guides (STIGs) Library	DISA STIGs Library
DISA Security Technical Implementation Guides	DISA STIGS

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Marine Corps Enterprise Cybersecurity Manual Library	USMC ECSM Library
MASTERING MICROSOFT EXCHANGE SERVER 2016	ISBN: 978-1-119-23205-6
MCSA WINDOWS 2016 COMPLETE STUDY GUIDE	ISBN: 978-1-119-35914-2
Microsoft Docs	https://docs.microsoft.com
Security Center Administrators Guide	http://static.tenable.com/prod_docs/SecurityCenter_5.0_Admin_Guide.pdf

PERSONNEL REQUIRED: 3 facilitators; 1 Course Chief, 1 Curriculum Developer

FACILITIES REQUIRED: Bldg 1763 Room 250

MATERIALS/EQUIPMENT:

Presentation Software PS	Instructors Intelligence Workstation (I) IW
Student Intelligence Workstation (S) IW	Admin Time AD
Check-in CHKIN	Checkout CKOUT
Computer Workstation w/ Software & Network Access CWS(N)	Dry Erase Board DB
End of Course Critique EOCC	Examination EXAM
Graduation Certificate Grad Cert	Learning Management System LMS
Lesson Plan LP	Laptop Computer LTC
Performance Evaluation Checklist PECL	Performance Exam PERF EXAM
PowerPoint Presentation PPT	Reference Materials (Digital and/or Paper Based) REF
Smart Board SB	Student Handout SH
Tablet TAB	Training Schedule TS

LEARNING EXPERIENCE ACTIVITIES:

Method	**Code**
Academic Orientation	ACOR
Administrative	ADMIN
Coaching	CCH
Course Introduction	CI
Check-in	CKIN
Check-Out	CKOUT
Demonstration	D
End of Course Critique	EOCC
Guided Discussion	GD
Graduation	GRAD
Gallery Walk	GW
Homework	HW
Informal Lecture	L
Informal Lecture/Guided Discussion	IL/GD
Informal Lecture/Gamification	IL/GM
Informal Lecture/Jigsaw	IL/JSW
Informal Lecture/Questioning	IL/Q
Independent Study Time	IST

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Leadership Guided Discussion	LDGD
Practical Application	PA
Quiz	Q
Socratic Questioning	SOQ
Teach Back	TB
Think-Pair-Share	TPS
Performance Exam	X(P)
Exam Review	X(RE)
Written Examination	X(W)

Learner Role(s): **Data Systems Administrator**

The Data Systems Administrator is a critical role in the 21st century digital landscape, responsible for designing, implementing, and managing data systems that support organizational operations. This role requires a deep understanding of the fundamentals of information technology, as well as the ability to apply 21st century learning concepts such as critical thinking, problem-solving, and collaboration.

Key Responsibilities:

- Design and implement data systems that meet organizational needs and requirements.
- Manage and maintain data systems, including hardware, software, and networks.
- Troubleshoot and resolve data system issues, using critical thinking and problem-solving skills.
- Collaborate with stakeholders to identify and prioritize data system needs and requirements.
- Develop and implement data backup and recovery plans, ensuring data integrity and security.
- Stay up to date with emerging trends and technologies in data systems administration, applying 21st century learning concepts such as innovation, creativity, and adaptability.

21st Century Learning Concepts:

- Critical thinking: analyzing complex data system issues and developing effective solutions.
- Problem-solving: troubleshooting and resolving data system problems, using creative and innovative approaches
- Collaboration: working with stakeholders to identify and prioritize data system needs and requirements
- Communication: effectively communicating data system information and requirements to stakeholders
- Creativity: designing and implementing innovative data systems solutions that meet organizational needs
- Innovation: staying up-to-date with emerging trends and technologies in data systems administration, and applying them to improve organizational operations

Learning Objectives:

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- Understand the fundamentals of information technology, including data systems, networks, and cybersecurity.
- Analyze complex data system issues and develop effective solutions, using critical thinking and problem-solving skills.
- Design and implement data systems that meet organizational needs and requirements, using collaboration and communication skills.
- Troubleshoot and resolve data system issues, using creative and innovative approaches.
- Develop and implement data backup and recovery plans, ensuring data integrity and security.
- Stay up-to-date with emerging trends and technologies in data systems administration, applying 21st century learning concepts such as innovation, creativity, and adaptability.

Performance Tasks:

- Design and implement a data system for a fictional organization, using critical thinking and problem-solving skills.
- Troubleshoot and resolve a complex data system issue, using creative and innovative approaches.
- Develop and implement a data backup and recovery plan, ensuring data integrity and security.
- Collaborate with stakeholders to identify and prioritize data system needs and requirements, using communication and collaboration skills.
- Create a presentation or report on emerging trends and technologies in data systems administration, applying 21st century learning concepts such as innovation, creativity, and adaptability.

Resources:

- Online learning platforms and resources, such as video tutorials and interactive simulations
- Textbooks and e-books on data systems administration and information technology
- Industry publications and research articles on emerging trends and technologies in data systems administration
- Collaboration tools, such as discussion boards and group projects, to facilitate communication and teamwork.

Facilitator Role: The Data Systems Administration Instructor plays a crucial role in delivering instruction and guiding learners in developing knowledge and skills related to data systems administration, including hardware, software, networks, and cybersecurity. To achieve this, the instructor employs a range of instructional methods, including lectures, discussions, and hands-on activities, to engage learners and promote understanding. By facilitating group discussions, providing individualized support, and using technology such as virtual labs and online resources, the instructor creates a positive and inclusive learning environment that encourages active learning and participation. The instructor's key

responsibilities include delivering instruction, guiding learners in completing performance tasks and projects, and evaluating learner progress, all while ensuring learners understand the fundamentals of information technology and can apply their knowledge to troubleshoot and resolve data system issues. Ultimately, the instructor's effectiveness is measured by learner engagement, understanding, and application of data systems administration concepts, as well as their ability to adapt instruction to meet learner needs and promote understanding.

LEARNING EXPERIENCE REFLECTION: Describe your reflection method. What feedback will you provide?

As I reflect on my learning experience, I will provide feedback on the following aspects:

- The effectiveness of the instructional methods and materials used in the course.
- The clarity and relevance of the learning objectives and outcomes
- The level of support and guidance provided by the instructor and peers.
- The opportunities for practice and application of the concepts and skills learned.
- The overall quality of the learning experience and its impact on my knowledge and skills in Data Systems Administration

I will provide constructive feedback that is specific, timely, and respectful, with the goal of improving the learning experience for future learners. My feedback will include suggestions for improvements, such as additional resources or support, and recommendations for how to build on the strengths of the course. By providing thoughtful and reflective feedback, I aim to contribute to the ongoing improvement and refinement of the Data Systems Administration course, and to help ensure that future learners have a high-quality and effective learning experience.

ASSESSMENT: How do learners demonstrate their understanding?

- Quizzes and exams to assess understanding of the fundamentals of information technology.
- Performance tasks and projects to assess application of critical thinking, problem-solving, and collaboration skills.
- Peer review and feedback to assess communication and collaboration skills.

SAFETY:

☐ Review/Reassess the learning environment/location for changes that affect the original Risk Assessment.

APPROVING SIGNATURE _____DATE _____

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